

SALAMANCA AP, Spain

WMO: 082020

Lat: 40.9596N

Lon: 5.4982W

Elev: 2605

StdP: 13.36

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.4	24.3	16.6	14.0	29.7	19.6	16.2	31.3	29.9	48.9	26.6	47.7	1.6	70	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.5	93.7	65.5	91.1	64.7	88.0	63.5	67.8	88.6	66.3	86.4	64.9	84.2	7.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	88.5	72.2	59.4	83.3	70.7	58.2	79.8	69.6	33.6	88.2	32.4	86.4	31.3	84.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.7	17.9	16.8	99.2	2.6	3.2	14.9	101.5	13.4	103.3	12.0	105.1	10.1	107.4
			16.1	71.4	2.7	2.8	14.2	73.4	12.6	75.0	11.1	76.6	9.1	78.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	40.0	41.9	47.0	51.2	58.3	66.5	70.5	70.1	63.5	55.8	45.5	40.7	
	DBStd	12.36	5.82	5.46	5.40	5.76	6.37	6.21	4.92	4.54	5.22	5.58	6.11	5.94	
	HDD50	1193	313	229	121	54	8	0	0	0	15	161	292		
	HDD65	4408	776	646	558	414	222	57	12	10	86	289	585	753	
	CDD50	2773	2	3	28	90	267	496	636	622	405	194	26	4	
	CDD65	513	0	0	0	0	15	103	183	167	42	3	0	0	
	CDH74	8357	0	0	2	47	379	1671	2789	2456	877	136	0	0	
	CDH80	3569	0	0	0	2	99	700	1331	1133	281	23	0	0	
Wind	WSAvg	6.5	6.8	7.4	8.0	8.2	7.1	6.5	5.5	5.3	5.1	5.6	6.8	6.3	
Precipitation	PrecAvg	16.5	1.7	1.2	1.2	1.7	1.8	0.9	0.4	0.5	1.2	2.5	1.9	1.6	
	PrecMax	24.8	4.9	3.1	4.3	4.4	4.5	2.4	1.5	1.3	3.0	5.9	7.8	4.4	
	PrecMin	10.8	0.0	0.1	0.0	0.3	0.2	0.0	0.0	0.0	0.0	0.5	0.2	0.2	
	PrecStd	3.2	1.2	0.9	0.9	1.0	1.1	0.7	0.4	0.4	0.7	1.5	1.5	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.5	65.1	72.2	78.7	87.8	95.8	97.2	97.1	91.9	83.2	68.2	59.5	
		MCWB	49.1	48.9	51.5	57.3	62.5	66.5	66.1	66.8	64.9	58.6	55.1	50.7	
	2%	DB	55.0	60.7	68.5	74.8	82.5	91.3	93.7	93.5	86.4	78.0	62.8	56.0	
		MCWB	48.8	48.2	50.3	55.3	60.4	65.1	65.3	65.5	62.1	57.3	52.8	49.8	
	5%	DB	52.1	56.9	64.8	70.1	78.2	87.7	91.0	90.2	82.8	73.2	59.1	53.5	
		MCWB	47.5	47.5	49.6	53.7	58.4	63.4	64.7	64.4	60.8	56.6	51.7	48.7	
	10%	DB	49.9	53.2	60.7	65.4	73.6	84.0	87.9	86.5	79.3	68.8	56.0	50.9	
		MCWB	46.0	46.2	48.7	52.1	56.9	62.1	63.8	63.2	60.0	55.9	50.5	47.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	52.6	55.0	59.3	65.0	69.0	69.7	69.7	67.0	62.8	57.6	53.6
			MCDB	55.4	58.6	65.2	74.1	82.8	88.7	90.8	91.8	87.0	73.4	63.7	56.5
2%		WB	50.2	50.4	52.9	56.8	61.8	66.6	67.5	67.5	64.1	60.3	55.1	51.2	
		MCDB	53.5	57.3	63.0	71.3	78.4	86.5	88.6	88.0	80.9	70.5	60.0	54.8	
5%		WB	48.3	48.5	51.3	54.8	59.9	64.8	66.1	65.9	62.7	58.9	53.2	49.4	
		MCDB	51.4	55.0	61.2	67.4	74.4	84.1	87.0	85.4	78.3	67.8	57.4	52.9	
10%		WB	46.4	46.7	49.5	52.8	58.0	63.1	64.8	64.5	61.2	57.4	51.4	47.5	
		MCDB	49.4	52.5	58.5	63.4	70.9	80.8	84.4	83.4	75.7	65.9	55.4	50.8	
Mean Daily Temperature Range	5% DB	MDBR	16.7	20.8	23.3	22.4	24.0	28.7	31.5	31.3	28.3	24.0	18.6	17.5	
		MCDBR	19.8	28.4	33.4	32.5	32.7	35.3	36.8	37.6	35.7	33.3	24.2	19.7	
		MCWBR	14.0	17.4	17.8	15.6	13.6	13.5	13.4	13.7	14.4	15.7	14.7	13.8	
	5% WB	MCDBR	16.1	23.5	26.1	27.8	28.4	32.4	34.3	33.9	31.3	24.0	18.5	16.6	
		MCWBR	12.3	15.3	14.9	14.2	12.8	12.9	13.2	13.3	13.5	12.7	12.5	12.5	
Clear-Sky Solar Irradiance	taub		0.281	0.293	0.326	0.334	0.345	0.349	0.334	0.342	0.329	0.319	0.294	0.279	
	taud		2.515	2.483	2.410	2.398	2.394	2.397	2.409	2.385	2.439	2.486	2.516	2.522	
	Ebn at Noon		282	294	294	298	295	293	296	291	289	280	274	273	
	Edh at Noon		23	28	33	36	37	37	36	36	32	27	23	21	
All-Sky Solar Radiation	RadAvg		623	945	1327	1700	2062	2356	2455	2135	1636	1078	697	557	
	RadStd		76	108	152	174	191	108	87	91	87	104	93	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.71	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	+1.17	N/A	N/A	N/A	N/A	N/A	+103

Nomenclature: See separate page